FINANCIAL INCLUSION AND FINANCIAL HEALTH IN KENYA

Socioeconomic Disparities in the 2024 FinAccess Household Survey

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Independent Research

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Evidence from the nationally representative 2024 FinAccess Household Survey

A cross-sectional, survey-weighted analysis of Kenyan adults aged 18 and above

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Abstract

Financial inclusion has expanded across Kenya, yet broad access to financial services does not necessarily indicate financial security. This study examines the distribution of financial access and financial health among Kenyan adults using the 2024 FinAccess Household Survey. The analysis draws on 19,750 respondents aged 18 years and above, uses individual survey weights, and provisionally accounts for 92 county-by-residence strata and 1,849 observed enumeration areas. National weighted estimates show that 84.83% of adults had formal financial access, whereas only 18.32% met the survey's financial-health criteria. Financial health ranged from 3.79% in the lowest household wealth quintile to 40.32% in the highest. A survey-weighted logistic regression of 19,702 complete cases found higher adjusted odds of financial health among formally included than financially excluded adults (adjusted odds ratio = 1.81, 95% confidence interval [1.13, 2.90]). Model-standardized predictions differed by 6.33 percentage points (provisional 95% confidence interval [1.89, 10.76]). Descriptive inequalities were also evident by education, livelihood, sex and residence. Results represent associations rather than causal effects; survey-design identifiers have not been independently certified, and adjusted-prediction intervals condition on the observed covariate distribution. The findings support financial sector monitoring that tracks financial health and resilience alongside access, rather than treating account inclusion as a sufficient measure of economic wellbeing.

Keywords: FinAccess 2024; Kenya; financial inclusion; financial health; survey-weighted regression; socioeconomic inequality

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1. Introduction

Kenya has become an important setting for the study of financial inclusion because formal financial services now reach a large share of the adult population. Digital payments, mobile money and regulated financial providers have changed how people receive, save and transfer money. Yet participation in a financial service does not by itself indicate whether a household can pay routine expenses, handle emergencies or invest for the future. The 2024 FinAccess Household Survey makes this distinction particularly visible: its official report documents formal financial access of 84.8%, while only 18.3% of adults meet the financial-health criteria (Central Bank of Kenya [CBK], Kenya National Bureau of Statistics [KNBS], & Financial Sector Deepening Kenya [FSD Kenya], 2024). The difference is not a causal paradox; the indicators measure separate constructs. A rigorous assessment of Kenya's financial sector should therefore report access and wellbeing separately while examining the socioeconomic context in which they occur.

International research similarly distinguishes ownership and use of financial services from economic security. The World Bank (2025) tracks financial access, payments, savings and financial resilience as related, but nonidentical, outcomes. The Consumer Financial Protection Bureau (2015) frames financial wellbeing around day-to-day control, capacity to absorb shocks, progress towards goals and freedom to make life choices. Kenyan evidence on mobile money adds an important, but narrower, perspective: Suri and Jack (2016) identified long-run welfare effects of mobile-money expansion in a different research design and period. Those findings cannot be transferred directly to an observational comparison of adults with different access classifications in 2024. In the present study, the research question concerns contemporary patterns and adjusted statistical associations, not the effect of introducing a new account or payment system. This distinction protects the analysis from treating access as an intervention when the survey did not assign it.

The study addresses three questions. First, how are formal inclusion, informal-only access, financial exclusion and financial health distributed among adults in 2024? Second, how do these outcomes vary across sex, age, rural or urban residence, education, livelihood and relative household wealth? Third, is formal financial access associated with financial health after adjustment for the measured characteristics? National estimates establish the scale of the outcomes; subgroup estimates identify concentrations of disadvantage; and regression models summarize conditional associations. The manuscript uses the official access and financial-health indicators, preserves the underlying survey weights and records each analytical exclusion. It does not redefine the supplied vulnerability index as a poverty measure because the index formula has not been independently documented. The contribution lies in combining reproducible descriptive analysis, exploratory design-based inference and transparent interpretation within the limits of the available public-use data.

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2. Literature Review

2.1 Financial inclusion and financial wellbeing

Financial inclusion commonly refers to access to and use of appropriate formal financial services, but the underlying objectives extend beyond opening accounts. The Global Findex Database provides comparative information on saving, borrowing, payments and financial resilience, making clear that financial behaviour and response to shocks deserve attention alongside access (Demirgüç-Kunt et al., 2022; World Bank, 2025). Financial wellbeing addresses a related but distinct dimension: whether people can meet obligations, protect themselves from adverse events and pursue future goals. The Consumer Financial Protection Bureau (2015) developed a broader consumer-centred framework that emphasizes control and security in the present and future. The 2024 FinAccess Household Survey adopts its own operational definitions, focusing on managing day-to-day needs, coping with risk and investing in future livelihoods (CBK et al., 2024). These frameworks are conceptually compatible, but they do not use identical questionnaires or classification rules. Accordingly, this study reports the official FinAccess health variable without presenting it as equivalent to another country's financial-wellbeing score.

2.2 What Kenyan evidence contributes

Research on Kenya illustrates both the promise of financial technology and the danger of overgeneralizing across research designs. Suri and Jack (2016) studied the expansion of mobile-money access and reported long-run effects on consumption and poverty, with important gender-related findings. Their work is influential because it links a specific financial technology to economic outcomes over time. However, their treatment, identification strategy, geographic setting and historical period differ from the variables available in the 2024 FinAccess cross-section. A respondent classified as formally included may use several kinds of services, and the classification itself is not equivalent to an experimentally assigned change in mobile-money access. The 2024 survey's descriptive evidence also shows substantial differences across rural and urban households, livelihood groups and educational levels (CBK et al., 2024). The present analysis extends that description by examining selected interactions and by estimating whether the observed access–health association remains after measured socioeconomic factors enter the model.

2.3 Why socioeconomic adjustment matters

Education, livelihood and asset ownership may be related both to financial access and to the capacity to cope with expenditure shocks. Consequently, an unadjusted difference between formally included and excluded adults may partly reflect differences in the groups' underlying circumstances. Adjusted regression can describe conditional associations, but it cannot eliminate bias from unobserved income volatility, household responsibilities, social networks or other relevant factors. Financial health may also shape financial service selection, which makes the

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direction of association ambiguous in a single cross-sectional survey. The wealth measure requires particular care because the FinAccess report constructs relative quintiles from household assets and living conditions using principal components, including separate rural and urban adjustments (CBK et al., 2024, Appendix 1). Asset quintiles are therefore not measures of monthly cash income. The interpretation in this manuscript treats wealth as a socioeconomic covariate rather than an exact proxy for earnings or a proven causal pathway.

2.4 Evidence gap and analytical contribution

The official survey already provides authoritative national indicators, so reproducing an 84.8% inclusion estimate is not itself a novel discovery. The added value of this manuscript is the transparent comparison of access, health and socioeconomic circumstances using one consistent adult population and a reproducible analytical workflow. The study examines whether the descriptive sex and urban–rural differences remain after adjustment and expresses the main health association in both odds ratios and standardized predicted probabilities. It also reports where inference is more uncertain: the public-use variables appear consistent with the official 92-stratum design but the original sampling identifiers have not been independently certified, and the financial vulnerability index lacks a confirmed derivation. Survey-analysis guidance emphasizes the importance of accounting for sample structure when estimating standard errors (Lumley, 2004). These features make the analysis useful as a carefully qualified secondary study, rather than a replacement for the official survey report or a causal evaluation of financial inclusion programmes.

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3. Data and Methods

3.1 Survey source and study population

The analysis uses the 2024 FinAccess Household Survey public-use dataset alongside its questionnaire and official report. The study selected enumeration areas, households and one eligible respondent per participating household through a multistage survey procedure. The fieldwork allowed participants aged 16 years and above, whereas the report's principal financial inclusion results concern adults aged 18 and above. The downloaded SPSS file contains 20,871 completed respondent records and 3,816 variables. The validated adult filter selects 19,750 respondents for the main descriptive estimates. Four adults have exact-age responses coded as unknown or refused, but their adult classification remains valid for analyses that do not use exact age. The research preserves all observations in the raw data, creates separate processed datasets and documents exclusions in script-generated audit reports. This approach makes the sample definition explicit and allows another analyst with authorized access to the microdata to retrace the preparation steps (CBK et al., 2024).

3.2 Measures and operational definitions

The analysis distinguishes three exclusive access categories: formally included, informal-only and excluded. The binary exclusion outcome equals one for adults in the excluded category. The main financial-health outcome is the official binary variable, finhealth_fnl, which indicates whether the survey's financial-health criteria are met. FinAccess measures financial health through abilities to manage routine expenses, cope with financial risks and invest in future needs; Appendix 3 of the official report identifies the questionnaire items used within those dimensions (CBK et al., 2024). The analysis does not apply a new cutoff to the available component scores because their exact original threshold rules have not been independently reconstructed. Covariates are sex, rural or urban residence, five adult age categories, education, principal livelihood and asset-based household wealth quintile. The documented vulnerability index remains outside the main regression because its formula and possible overlap with health components have not been independently established.

3.3 Survey weights and estimation strategy

Individual survey weights reproduced official formal-access, exclusion and financial-health rates to within 0.04 percentage points. The report describes 92 strata: rural and urban in 45 counties, and urban only in Nairobi and Mombasa (CBK et al., 2024). DataFirst (2026) separately constructed the same 92 county-by-residence strata in its curated release. Here, county and residence form working strata, while 1,849 observed cluster identifiers serve as candidate primary sampling units. No stratum contains a single observed cluster. The official survey planned 1,885 enumeration areas (EAs). The public-use records contain 36 fewer observed clusters, concentrated

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in Trans Nzoia (16) and Garissa (9). The report documents 20 selected EAs that could not be covered but does not account for the complete observed difference. The published Appendix 4 urban grand total (705) conflicts with the sum of county rows (728); the county values were used for comparison. Taylor-linearized standard errors follow the candidate design, while all intervals and tests remain provisional because the original PSU identifiers have not been independently certified (Lumley, 2004).

3.4 Regression, marginal predictions and sample exclusions

Two survey-weighted logistic models examine financial exclusion and financial health. The financial-exclusion model adjusts for sex, residence, age category, education, livelihood and wealth quintile. The financial-health model adds financial access status to the same adjustment set, with financially excluded adults serving as the reference group. The models use quasibinomial estimation and present adjusted odds ratios rather than causal risk ratios. Model-standardized probabilities average predicted values for alternative access categories across the same observed complete-case covariate distribution. Their reported delta-method intervals use the coefficient covariance matrix and treat that covariate distribution as fixed, making them narrower in scope than fully design-based marginal-standardization intervals. The 19,702-person complete-case sample excludes four adults with unavailable exact-age groups, eight in the small Other education category and 36 lacking valid wealth quintiles. These cases remain in analyses that do not require the affected predictors; no missing wealth values were imputed.

3.5 Reproducibility and quality assurance

Interview IDs, individual weights, inclusion and health classifications, and cluster-to-stratum consistency passed automated checks. The 29-variable analytical dataset maintained official national benchmarks; 36 wealth-quintile values remain missing and questionnaire skip patterns were preserved. All 38 regression coefficients reproduced on rerun, and the models converged. A supplementary county audit identified 28 rural and eight urban planned EAs without observed respondent records. Every county-by-residence observed cluster count was at or below its planned allocation. For national indicators, replacing the candidate stratified-cluster design with a cluster-only model increased standard errors (percentage points) from 0.403 to 0.456 for formal access, 0.342 to 0.371 for exclusion, and 0.536 to 0.623 for financial health. Point estimates remained fixed because the same weights and observations were used. An independent-weighted alternative, which ignores clustering, was examined only as a sensitivity diagnostic. These checks cannot certify original PSU identifiers, rule out model confounding, or establish sensitivity of regression standard errors under alternative designs.

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4. Results

4.1 National access and financial health

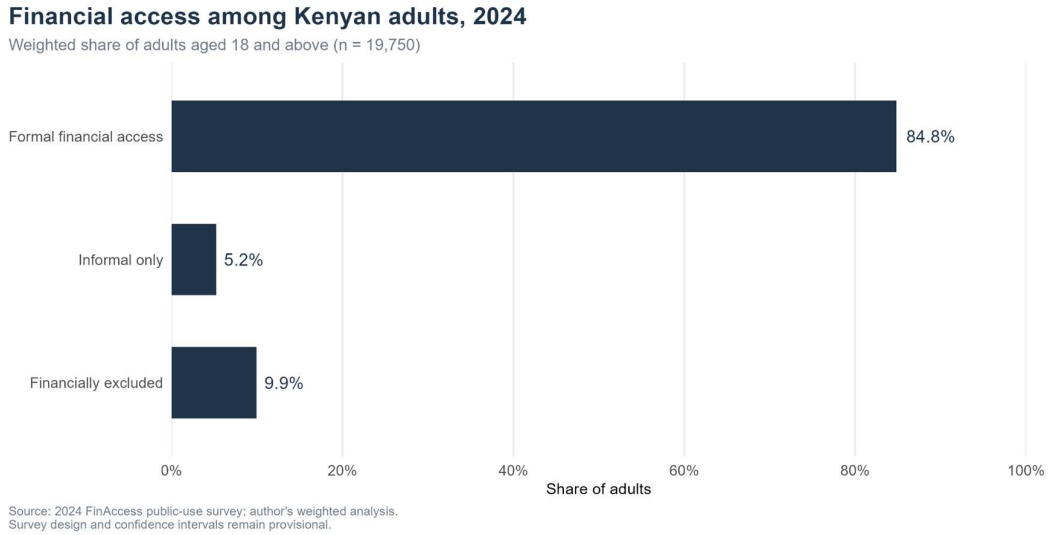
Financial access and financial health differ sharply at the national level. An estimated 84.83% of adults meet the survey's formal financial-access definition, 5.23% report informal-only access and 9.94% are classified as excluded. Only 18.32% meet the financial-health criteria. The formal, informal-only and excluded groups are mutually exclusive and together account for approximately 100% of adults, allowing for rounding. In contrast, financial health is a separate binary classification that may occur within any access category. The gap between the national percentages must therefore not be interpreted as the proportion of formally included adults who are unhealthy. Table 1 summarizes the weighted estimates and provisional confidence intervals, while Figure 1 displays the three access categories. These estimates closely reproduce the national benchmarks in the official report (CBK et al., 2024), providing a practical validity check for the adult filter and individual weights.

Table 1: National financial access and financial health among adults, 2024

Indicator	Unweighted n	Estimate (95% CI), %
Formal financial access	19750	84.83 [84.04, 85.62]
Informal-only access	19750	5.23 [4.81, 5.66]
Financial exclusion	19750	9.94 [9.27, 10.61]
Financially healthy	19750	18.32 [17.27, 19.38]

Note. Individual-weighted percentages among adults aged 18 and above (n = 19,750). Confidence intervals are provisional pending independent certification of the original cluster and stratum identifiers. Source: author's analysis of the 2024 FinAccess public-use dataset.

Figure 1: Financial access among Kenyan adults, 2024



Note. The bars show survey-weighted shares. Source: 2024 FinAccess public-use data; author's analysis. Survey-design uncertainty remains provisional.

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4.2 Financial health within access categories

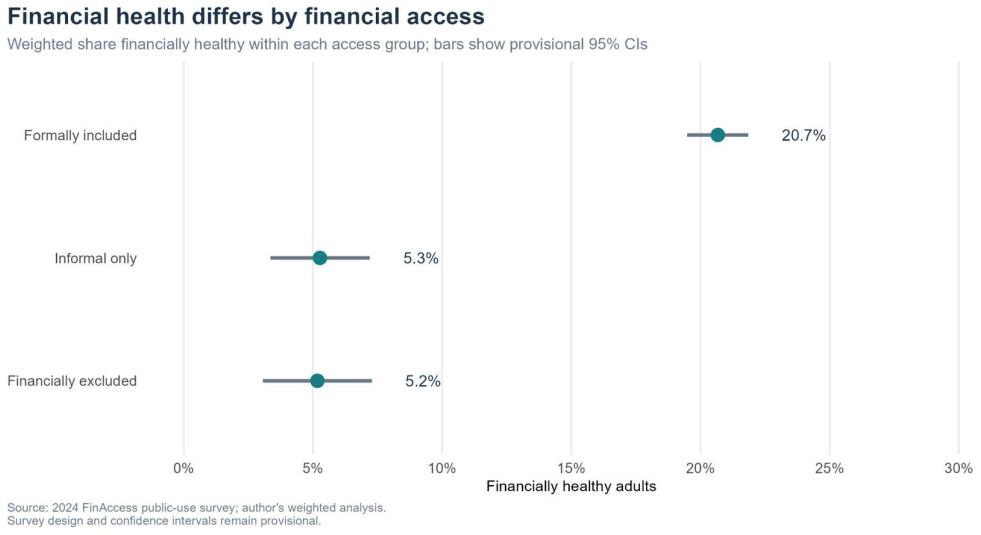
Financial health is substantially more common among formally included adults than among the other two access groups. The weighted health estimate is 20.67% for the formally included, 5.27% for informal-only respondents and 5.17% for those financially excluded. The corresponding unweighted group sizes are 16,923, 1,068 and 1,759. These estimates replicate the rounded health-by-access figures in the official report (CBK et al., 2024). The similarity between the informal-only and excluded point estimates does not prove equivalence, particularly given the wider intervals around the smaller subgroups. Table 2 and Figure 2 make the within-group denominators explicit. Most formally included adults still fail to meet the health threshold, which strengthens the descriptive case for monitoring financial health as its own outcome. The observed difference may nevertheless reflect the groups' education, livelihoods, wealth and other measured or unmeasured circumstances.

Table 2:Financial health by financial-access classification

Financial access	Unweighted n	Financially healthy, % (95% CI)
Formally included	16,923	20.67 [19.48, 21.85]
Informal-only	1,068	5.27 [3.35, 7.20]
Financially excluded	1,759	5.17 [3.06, 7.28]

Note. Weighted prevalence within each access category. Confidence intervals are provisional; groups are not experimental treatment groups.

Figure 2: Financial health by financial-access classification



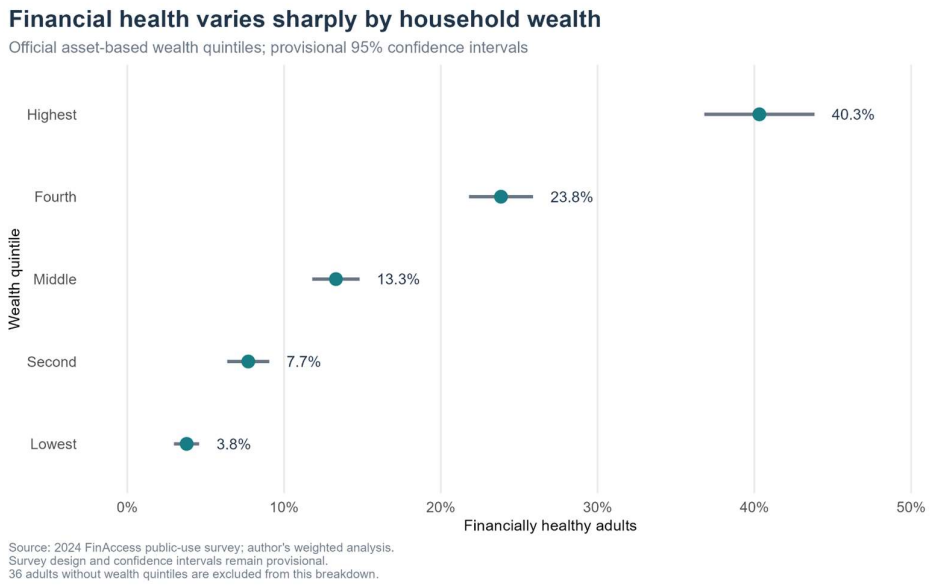
Note. Percentages describe health within access categories; horizontal bars show provisional 95% confidence intervals. Source: 2024 FinAccess; author's analysis.

4.3 Socioeconomic inequalities

The strongest observed gradient in financial health appears across relative household wealth. The proportion classified as healthy rises from 3.79% in the lowest quintile to 7.72% in the second,

13.31% in the middle, 23.84% in the fourth and 40.32% in the highest. The highest-minus-lowest descriptive difference is 36.53 percentage points using rounded subgroup estimates; the unrounded survey contrast yields 36.54 percentage points. These are quintiles of an asset-based relative wealth index, not measured income brackets. Education and livelihood reveal similarly pronounced patterns: health is 3.38% among adults with no education and 43.46% among those with tertiary education, while it is 8.61% among casual workers and 45.18% among employed adults. Figure 3 and Table 3 present the estimates without attributing the gaps to a single cause. A person's education, livelihood and asset ownership may jointly shape financial circumstances.

Figure 3: Financial health across relative household wealth quintiles



Note. Wealth is an asset-based index described in the official FinAccess report (Appendix 1), not a direct income measure. Thirty-six adults with missing quintiles are excluded from these subgroup estimates.

Table 3: Selected descriptive differences in financial inclusion and financial health

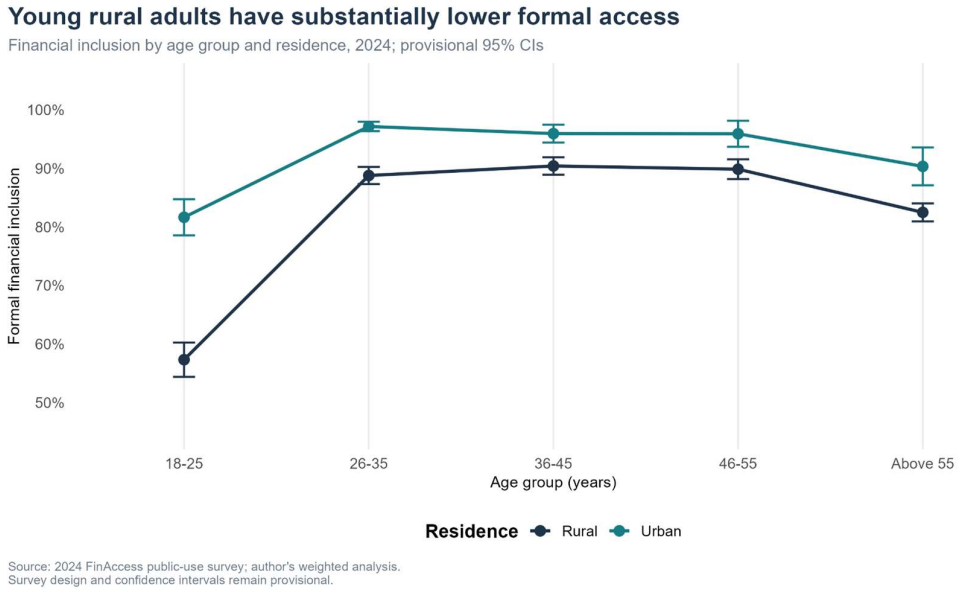
Population characteristic	**Category**	**Formal access, %**	**Financial health, %**
Residence	Rural	80.18	12.63
Residence	Urban	91.34	26.29
Sex	Female	84.03	14.68
Sex	Male	85.68	22.19
Age	18–25	69.04	14.89
Age	26–35	92.95	23.92
Education	No formal education	75.37	3.38
Education	Tertiary	98.48	43.46
Livelihood	Casual worker	83.35	8.61
Livelihood	Employed	96.88	45.18
Wealth	Lowest	70.18	3.79
Wealth	Highest	93.62	40.32

Note. Values are individual-weighted descriptive percentages; complete category distributions and provisional confidence intervals appear in Appendix Table A1. The education category Other (n = 8) is suppressed in detailed subgroup reporting.

4.4 Age, residence and sex intersections

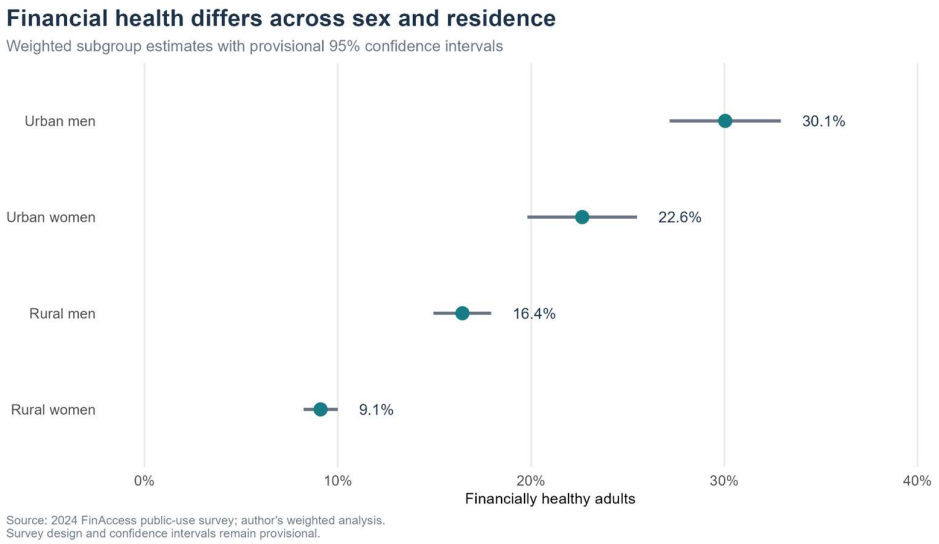
Combining age with residence reveals that formal access is particularly low among rural adults aged 18–25. In this group, the estimated access rate is 57.34%, compared with 81.65% among urban adults of the same age. Adults aged 26–35 show substantially higher rates in both rural (88.78%) and urban (97.16%) areas. These figures identify distinct subpopulations for further assessment, but the plot connects age categories within one 2024 cross-section, not repeated observations of the same respondents over time. Sex-by-residence analysis highlights a different inequality: 9.11% of rural women meet the financial-health criteria, compared with 16.45% of rural men, 22.65% of urban women and 30.05% of urban men. The intersectional results are descriptive; they do not isolate independent effects of sex or residence and may reflect the joint distribution of education, wealth, livelihood and other characteristics.

Figure 4: Formal financial access by age group and residence



Note. Age groups represent separate respondents in the same 2024 survey, not trends over time. The intervals and design remain provisional.

Figure 5: Financial health by sex and residence



Note. Weighted subgroup estimates with provisional 95% confidence intervals; no causal or interaction effects are inferred from this descriptive figure.

4.5 Regression results and conditional associations

The financial-exclusion model shows that adults aged 26–35 have markedly lower adjusted odds of exclusion than those aged 18–25 (adjusted odds ratio [AOR] = 0.12, provisional 95% CI [0.10, 0.15]). Relative to adults with no formal education, those with tertiary education have lower adjusted odds of exclusion (AOR = 0.06, 95% CI [0.03, 0.15]). Dependents have higher adjusted odds than casual workers (AOR = 2.35, 95% CI [1.91, 2.88]). In the financial-health model, formally included adults have higher adjusted odds of financial health than excluded adults after controlling for the specified covariates (AOR = 1.81, 95% CI [1.13, 2.90]), while the informal-only contrast is imprecise (AOR = 1.10, 95% CI [0.59, 2.05]). Employment, education and wealth also show substantial adjusted associations. Odds ratios are not risk ratios or causal effects; Table 4 summarizes selected health model coefficients.

Table 4: Selected adjusted associations with financial health

Predictor (reference)	AOR (95% CI)	p
Formal access (excluded)	1.81 [1.13, 2.90]	.014
Informal-only (excluded)	1.10 [0.59, 2.05]	.754
Male (female)	1.56 [1.37, 1.79]	< .001
Urban (rural)	0.77 [0.64, 0.93]	.006
Tertiary education (none)	4.73 [3.43, 6.54]	< .001
Employed (casual worker)	4.11 [3.23, 5.23]	< .001
Highest wealth (lowest)	8.94 [6.45, 12.38]	< .001

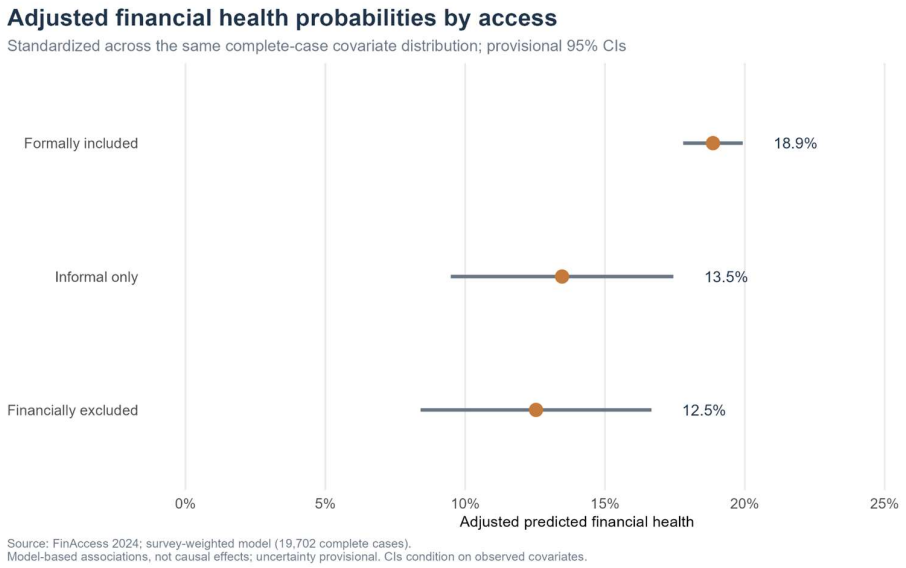
Note. AOR = adjusted odds ratio. The model includes age, education, livelihood, wealth, sex, residence and access classification (n = 19,702 complete cases). Intervals and tests remain exploratory and provisional. Full coefficients appear in Appendix Table A3.

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4.6 Adjusted probabilities and sensitivity

Model-standardized financial-health probabilities translate selected logistic coefficients into a more interpretable probability scale. The predicted prevalence of financial health is 18.86% under the formal-inclusion category, 13.47% under informal-only access and 12.53% under exclusion when each scenario is averaged across the same complete-case covariate distribution. The formal-inclusion minus excluded difference is 6.33 percentage points, with a provisional 95% interval of 1.89–10.76; the informal-only difference is 0.93 points, with an interval crossing zero. Figure 6 and Appendix Table A6 display the principal adjusted comparisons. These estimates describe fitted scenarios, not the causal effect of changing a person's financial access classification. Furthermore, the delta-method intervals treat the observed covariate distribution as fixed and omit part of the uncertainty associated with population standardization. Consequently, their interpretation must remain narrower than that of a fully design-based marginal estimate.

Figure 6: Model-standardized financial health by financial-access status



Note. Model-standardized probabilities for 19,702 complete cases; intervals condition on the observed covariate distribution and use provisional survey covariance estimates. Not causal effects.

5. Discussion

The study's clearest descriptive conclusion is that high formal financial access coexists with low financial health. Approximately 85% of adults satisfy the official formal-access definition, yet fewer than one in five meet the financial-health criteria. This is consistent with FinAccess's decision to distinguish access, usage, quality and impact rather than collapse financial inclusion into a single indicator (CBK et al., 2024). It also fits wider financial-wellbeing frameworks that examine people's capacity to handle routine expenditure, withstand setbacks and meet future goals (Consumer Financial Protection Bureau, 2015). The result does not show that formal services fail or succeed as an intervention; the survey captures observed status in one period. Instead, it indicates that a programme reporting only account inclusion could miss important differences in resilience and financial circumstances. A more informative monitoring approach would retain access alongside complementary indicators on coping capacity, basic needs and future-oriented savings or investment.

Relative household wealth, educational attainment and livelihood are closely associated with financial health in both descriptive and adjusted comparisons. The difference between the highest and lowest asset-based quintiles reaches more than 36 percentage points in the descriptive estimates, while employment and tertiary education also have large conditional odds ratios. These patterns make economic context central to interpreting financial inclusion. They should not be framed as evidence that a particular programme would produce the observed differences by moving respondents into a different wealth or employment category. The FinAccess wealth measure is built from asset and living-condition information, and groups can differ on multiple unmeasured dimensions. The World Bank (2025) similarly treats financial resilience and account use as distinct indicators, while earlier research in Kenya has linked particular changes in mobile-money access to welfare under different identification strategies (Suri & Jack, 2016). Our analysis is compatible with examining those mechanisms, but it does not identify them.

The shift in the urban–rural coefficient illustrates why conditional models require careful explanation. Urban adults have much higher observed financial-health prevalence, but the fully adjusted urban odds ratio is below one. Sequential models show that the sign changes when household wealth is added after access, sex, age, education and livelihood. This pattern is consistent with substantial differences in socioeconomic composition, although logistic odds ratios are also non-collapsible and the coefficient change cannot be assigned to a single causal pathway. The adjusted result should not be described as proof that living in a city damages financial health. A related pattern occurs in the exclusion model: the male coefficient changes after education and livelihood enter the specification. Readers should be given both the unadjusted rates and the adjusted estimates, with clear statements about the reference categories and denominators. Neither set invalidates the other; each answers a different descriptive question.

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6. Policy Implications

The findings support a measurement recommendation rather than an untested claim about programme effectiveness. Kenyan financial inclusion initiatives can continue to track the proportion of adults using formal financial services, but they should report financial health as a separate outcome. The 2024 FinAccess measurement framework provides a practical starting point through its three dimensions of daily money management, risk coping and future investment (CBK et al., 2024). Programme monitoring could use these dimensions to identify where formal access coexists with weak resilience, while documenting how indicators are collected and which populations are represented. Financial service providers could also assess product suitability, affordability and consumer protection rather than assuming that the presence of an account establishes meaningful welfare improvement. These are evidence-informed areas for further evaluation, not interventions whose effectiveness has been demonstrated by the present cross-sectional models.

The descriptive analysis identifies groups that may merit additional diagnostic research: rural adults aged 18–25 have notably low formal access, while rural women have the lowest financial-health rate among the reported sex-by-residence categories. These results support targeted investigation of barriers, financial needs and service use in those populations. They do not establish that the groups face the same underlying constraints or that one outreach method would work for all. Before implementing targeted strategies, programme teams should triangulate the survey patterns with qualitative interviews, geographic service coverage, affordability information and, where available, administrative evidence. Subsequent evaluations could establish baseline and follow-up measures that capture both use and quality of financial services. The recommendation is to test plausible programme responses, not to infer from observational associations that distributing accounts or loans would automatically improve household resilience.

For research organisations, the priority is clear documentation of the sampling structure, derived indicators and public-use variance-estimation assumptions. The present analysis reconstructed 92 county-by-residence strata in line with the official report and DataFirst metadata, examined 1,849 observed clusters against published county allocations, and compared three working variance designs. These checks improve transparency but leave original PSU certification unresolved. Future analysis can review additional public metadata or run prespecified alternatives if suitable documented identifiers become available. The public release should distinguish benchmark-matching point estimates from provisional tests and intervals and avoid unsupported interpretations of the undocumented vulnerability index.

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7. Limitations

The study is a secondary analysis of a cross-sectional survey and cannot determine whether formal financial access preceded or followed improvements in financial health. Respondents may select financial services partly because of their resources, opportunities and economic needs, and unobserved income volatility, household obligations or financial knowledge may affect both access and health. The models adjust for several measured characteristics, but residual confounding and reverse causation remain plausible. The financial-health variable reproduces the official national benchmark and its component scores are internally consistent, yet the analysis has not independently reconstructed every classification threshold from the original questionnaire. The public-use vulnerability index also lacks a fully verified formula, so it is not used as the central outcome or as an explanatory predictor of the financial-health classification. The interpretation should therefore remain specific to the official measures rather than claim to capture every dimension of poverty or economic security.

The provided individual weights reproduce official national inclusion and health benchmarks, and the reconstructed 92 county-by-residence strata agree with the published sampling description and the independent DataFirst (2026) archival reconstruction. Nevertheless, the original public-use PSU identifiers were not independently certified. The public-use data contain 1,849 observed clusters compared with 1,885 planned EAs. Only 20 EAs are explicitly identified as uncovered in the official report; the full 36-cluster difference remains unexplained. Of particular concern for local inference, Trans Nzoia has 16 fewer clusters than planned and Garissa has nine fewer. Appendix 4 also prints 705 as the urban-cluster grand total even though the county rows sum to 728. The alternative-design calculations retain identical national weighted point estimates, but their standard errors differ and do not validate the original sampling design. Consequently, all confidence intervals, regression tests and p-values are described as provisional; this sensitivity exercise covers only three national indicators, not the regression parameters. The model-standardized probability intervals condition on the observed complete-case covariate distribution and omit part of its estimation uncertainty. Forty-eight adults are excluded from complete-case regressions, and the seven post hoc comparisons receive a Holm adjustment only within that selected family (Holm, 1979). These limits should be retained in public figures, reporting and policy interpretation.

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8. Conclusion

The 2024 FinAccess Household Survey shows that formal financial access is widespread among Kenyan adults but financial health remains substantially lower and unevenly distributed. The weighted adult estimate of formal access is 84.83%, compared with 18.32% classified as financially healthy. Differences are particularly pronounced across relative household wealth, education, livelihood and residence, while the joint age-by-residence analysis identifies limited access among younger rural adults. Formal access remains positively associated with financial health after adjustment for measured socioeconomic factors, and the model-standardized comparison is 6.33 percentage points higher for the formal-inclusion scenario than for exclusion. These results are not causal and must be read alongside the unresolved survey-design qualifications. The central contribution is to demonstrate a reproducible approach to comparing access with financial wellbeing rather than treating one as a sufficient proxy for the other. The findings provide a useful foundation for research, financial-sector monitoring and more carefully evaluated programmes.

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Appendix: Supplementary Statistical Evidence

The following supplementary tables and figures preserve the supporting statistical results without crowding the main Results section. The appendix contains the full descriptive categories, exploratory comparison family and both fitted regression coefficient sets. Each table uses the same values as the scripted analyses. The numbers remain subject to the reconstructed 92-stratum and 1,849-cluster design, and the probability intervals treat the observed covariate distribution as fixed. The appendix should not be understood as independent validation of the original survey design; rather, it supplies a transparent record of the numerical evidence behind the main narrative. Variables and categories with very small samples were suppressed where indicated. These supporting results can be checked against the project's aggregate CSV outputs without distributing the respondent-level public-use records.

Table 5: Full demographic and socioeconomic subgroup estimates

Category (group)	n	Formal access % [CI]	Financial health % [CI]	Excluded % [CI]
Sex: Female	11739	84.03 [82.95, 85.11]	14.68 [13.35, 16.00]	10.04 [9.13, 10.94]
Sex: Male	8011	85.68 [84.54, 86.83]	22.19 [20.68, 23.71]	9.83 [8.83, 10.83]
Residence: Rural	12741	80.18 [79.12, 81.25]	12.63 [11.73, 13.54]	12.61 [11.71, 13.50]
Residence: Urban	7009	91.34 [90.24, 92.45]	26.29 [24.17, 28.41]	6.20 [5.21, 7.18]
Age group: 18-25	3997	69.04 [66.75, 71.34]	14.89 [13.13, 16.65]	23.14 [21.06, 25.23]
Age group: 26-35	5361	92.95 [92.08, 93.83]	23.92 [21.82, 26.01]	3.10 [2.55, 3.65]
Age group: 36-45	3942	92.65 [91.56, 93.75]	18.86 [16.63, 21.09]	4.09 [3.22, 4.97]
Age group: 46-55	2545	91.87 [90.49, 93.25]	18.03 [15.48, 20.57]	4.08 [3.12, 5.04]
Age group: Above 55	3901	84.10 [82.70, 85.51]	13.44 [11.18, 15.70]	9.70 [8.54, 10.86]
Education: None	3021	75.37 [72.96, 77.78]	3.38 [2.54, 4.22]	16.66 [14.56, 18.75]
Education: Primary	7369	81.85 [80.57, 83.12]	9.96 [8.70, 11.22]	10.43 [9.34, 11.52]
Education: Secondary	6599	83.78 [82.42, 85.15]	18.39 [16.64, 20.14]	11.84 [10.60, 13.08]
Education: Tertiary	2753	98.48 [97.65, 99.30]	43.46 [40.14, 46.78]	0.96 [0.21, 1.72]
Livelihood: Casual Worker	4784	83.35 [81.82, 84.88]	8.61 [7.25, 9.98]	9.47 [8.25, 10.70]
Livelihood: Dependent	4309	73.95 [71.66, 76.24]	12.24 [9.71, 14.76]	20.79 [18.70, 22.88]
Livelihood: Agriculture	4514	87.16 [85.79, 88.52]	15.64 [13.92, 17.36]	6.56 [5.54, 7.58]
Livelihood: Own Business	3280	92.30 [91.03, 93.57]	25.63 [23.04, 28.21]	3.89 [2.94, 4.84]
Livelihood: Employed	2169	96.88 [95.99, 97.78]	45.18 [41.78, 48.58]	2.00 [1.24, 2.76]
Livelihood: Other	694	71.38 [66.41, 76.36]	7.44 [3.60, 11.29]	21.25 [16.73, 25.77]
Wealth quintile: Lowest	4640	70.18 [68.25, 72.11]	3.79 [2.98, 4.59]	19.27 [17.57, 20.98]
Wealth quintile: Second	4179	79.55 [77.73, 81.37]	7.72 [6.38, 9.06]	12.64 [11.18, 14.09]
Wealth quintile: Middle	4061	86.96 [85.50, 88.41]	13.31 [11.80, 14.82]	8.10 [6.85, 9.35]
Wealth quintile: Fourth	4051	92.23 [90.88, 93.57]	23.84 [21.80, 25.89]	5.18 [3.96, 6.41]
Wealth quintile: Highest	2783	93.62 [91.94, 95.29]	40.32 [36.81, 43.84]	5.38 [3.82, 6.94]

Note. Each estimate is weighted; CIs are provisional. Source: Phase 3.1 subgroup output. Missing wealth values are excluded only from wealth rows. Education Other (n = 8) remains suppressed.

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Table 6: Seven post hoc comparisons with Holm adjustment

Contrast	Difference in percentage points (95% CI)	Holm-adjusted p
Sex: male vs female (formal access)	1.65 [0.09, 3.22]	.038
Sex: male vs female (financial health)	7.52 [5.60, 9.43]	< .001
Residence: urban vs rural (formal access)	11.16 [9.62, 12.69]	< .001
Residence: urban vs rural (financial health)	13.66 [11.35, 15.97]	< .001
Age: 26-35 vs 18-25 (financial exclusion)	-20.04 [-22.18, -17.91]	< .001
Wealth: highest vs lowest (financial health)	36.54 [32.93, 40.14]	< .001
Access: formal vs excluded (financial health)	15.50 [13.09, 17.91]	< .001

Note. Differences in percentage points; comparison minus reference group. Holm-adjusted p-values cover the seven exploratory tests only.

Table 7: Full survey-weighted logistic regression: financial health

Predictor	Reference	Adjusted OR (95% CI)	p-value
Financial access: Informal only	Financially excluded	1.10 [0.59, 2.05]	.754
Financial access: Formally included	Financially excluded	1.81 [1.13, 2.90]	.014
Sex: Male	Female	1.56 [1.37, 1.79]	< .001
Residence: Urban	Rural	0.77 [0.64, 0.93]	.006
Age: 26-35	18-25	1.32 [1.02, 1.71]	.037
Age: 36-45	18-25	1.25 [0.94, 1.66]	.121
Age: 46-55	18-25	1.20 [0.91, 1.59]	.199
Age: Above 55	18-25	1.32 [0.98, 1.79]	.070
Education: Primary	None	1.82 [1.33, 2.49]	< .001
Education: Secondary	None	2.61 [1.91, 3.57]	< .001
Education: Tertiary	None	4.73 [3.43, 6.54]	< .001
Livelihood: Dependent	Casual Worker	1.40 [1.02, 1.94]	.039
Livelihood: Other	Casual Worker	1.11 [0.63, 1.97]	.712
Livelihood: Agriculture	Casual Worker	2.64 [2.07, 3.36]	< .001
Livelihood: Own Business	Casual Worker	2.77 [2.18, 3.52]	< .001
Livelihood: Employed	Casual Worker	4.11 [3.23, 5.23]	< .001
Wealth: Second	Lowest	1.62 [1.21, 2.17]	.001
Wealth: Middle	Lowest	2.51 [1.91, 3.32]	< .001
Wealth: Fourth	Lowest	4.44 [3.31, 5.97]	< .001
Wealth: Highest	Lowest	8.94 [6.45, 12.38]	< .001

Note. Model n = 19,702; adjusted odds ratios with provisional 95% confidence intervals. All coefficients use the corresponding categories shown in the reference column.

Table 8: Full survey-weighted logistic regression: financial exclusion

Predictor	Reference	Adjusted OR (95% CI)	p-value
Sex: Male	Female	1.30 [1.10, 1.54]	.002
Residence: Urban	Rural	0.70 [0.54, 0.91]	.007
Age: 26-35	18-25	0.12 [0.10, 0.15]	< .001
Age: 36-45	18-25	0.14 [0.10, 0.18]	< .001
Age: 46-55	18-25	0.13 [0.10, 0.18]	< .001
Age: Above 55	18-25	0.19 [0.15, 0.24]	< .001
Education: Primary	None	0.88 [0.71, 1.10]	.258

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Predictor	Reference	Adjusted OR (95% CI)	p-value
Education: Secondary	None	0.59 [0.45, 0.76]	< .001
Education: Tertiary	None	0.06 [0.03, 0.15]	< .001
Livelihood: Dependent	Casual Worker	2.35 [1.91, 2.88]	< .001
Livelihood: Other	Casual Worker	1.86 [1.29, 2.70]	< .001
Livelihood: Agriculture	Casual Worker	0.77 [0.61, 0.97]	.024
Livelihood: Own Business	Casual Worker	0.62 [0.46, 0.85]	.003
Livelihood: Employed	Casual Worker	0.40 [0.26, 0.61]	< .001
Wealth: Second	Lowest	0.73 [0.60, 0.88]	.001
Wealth: Middle	Lowest	0.47 [0.37, 0.60]	< .001
Wealth: Fourth	Lowest	0.37 [0.26, 0.52]	< .001
Wealth: Highest	Lowest	0.50 [0.33, 0.76]	< .001

Note. Model n = 19,702; adjusted odds ratios with provisional 95% confidence intervals. Results are conditional associations, not treatment effects.

Table 9: Full model-standardized financial-health probabilities

Comparison	Category	Predicted healthy (95% CI), %
Financial access	Financially excluded	12.53 [8.41, 16.66]
Financial access	Informal only	13.47 [9.48, 17.45]
Financial access	Formally included	18.86 [17.80, 19.93]
Residence	Rural	20.16 [18.74, 21.58]
Residence	Urban	17.06 [15.61, 18.50]
Sex	Female	15.69 [14.44, 16.95]
Sex	Male	21.03 [19.74, 22.31]

Note. Marginal standardization across the same complete-case covariate distribution. Intervals are coefficient-based delta-method approximations and remain provisional.

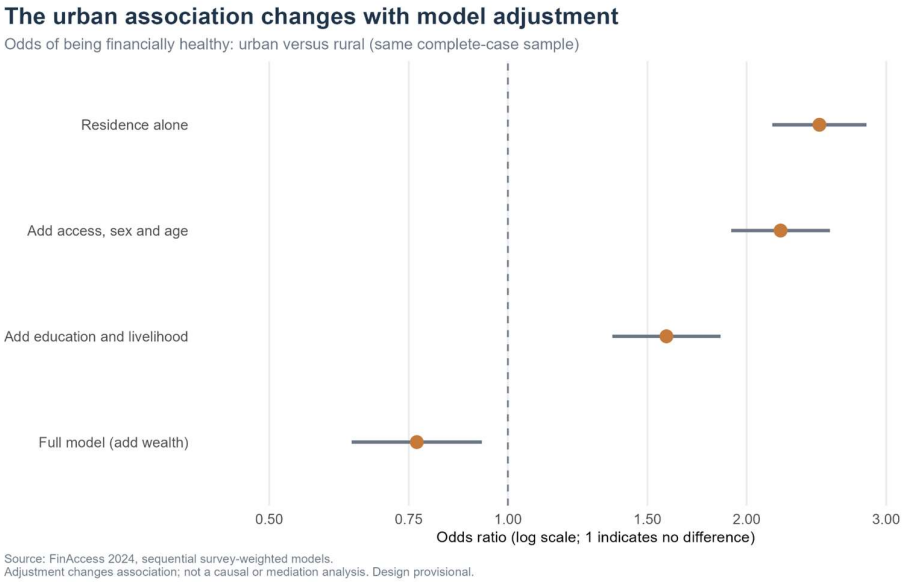
Table 10: Adjusted differences in predicted financial health

Comparison (vs. reference)	Difference, pp (95% CI)	p
Formally included vs. excluded	+6.33 [1.89, 10.76]	.005
Informal-only vs. excluded	+0.93 [−4.89, 6.75]	.753
Urban vs. rural	−3.10 [−5.26, −0.94]	.005
Male vs. female	+5.33 [3.73, 6.94]	< .001

Note. pp = percentage points. Exploratory p-values are not Holm-adjusted. These probability differences use coefficient-covariance delta-method intervals and cannot be interpreted as treatment effects.

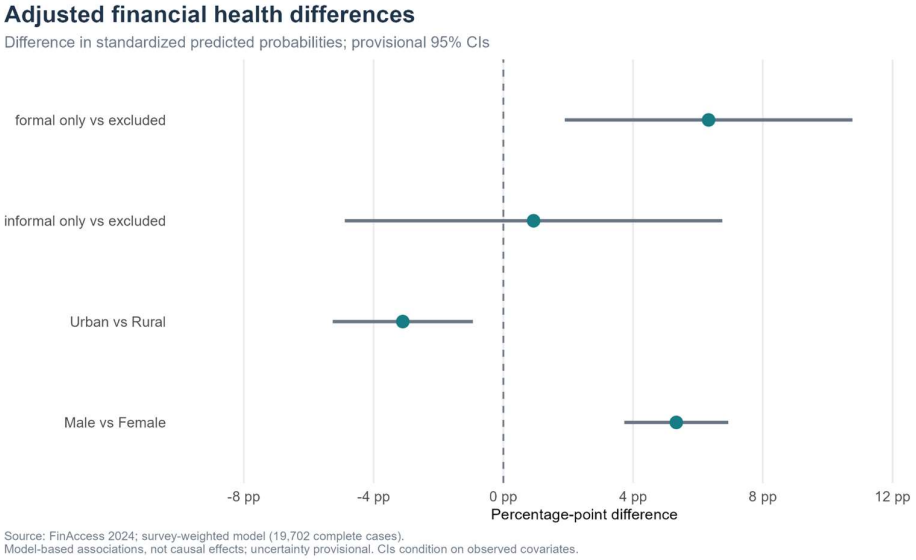
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Figure 7: Urban–rural association across sequential regression specifications



Note. The same 19,702-person sample is used at every stage. Adjustment changes the estimated odds ratio; it does not identify a causal mechanism.

Figure 8: Adjusted financial-health probability differences



Note. Model-standardized differences in percentage points, with provisional delta-method intervals. Differences are not causal effects.